



Chapter-7 BINOMIAL THEOREM

PRACTICE SHEET

1. Sum of all odd position terms of the expansion $(1+x)^8$?
(a) 256 (b) 512
(c) 1024 (d) 128
2. What are the last two digits of the number 9^{200} ?
(a) 19 (b) 21
(c) 41 (d) 01
3. For any positive integer n , if $4^n - 3n$ is divided by 9, then what is the remainder?
(a) 8 (b) 6
(c) 4 (d) 1
4. What is the coefficient of x^5 in the expansion $(1 - 2x + 3x^2 - 4x^3 + \dots + \infty)^{-5}$?
(a) $(10!)/(5!)^2$ (b) 5^{-5}
(c) 55 (d) $10!/\{6!(4!)\}$
5. What is the middle term in the expansion of $\left(\frac{x\sqrt{y}}{3} - \frac{3}{y\sqrt{x}}\right)^{12}$
(a) $C(12, 7) x^3 y^{-3}$ (b) $C(12, 6) x^{-3} y^3$
(c) $C(12, 7) x^{-3} y^3$ (d) $C(12, 6) x^3 y^{-3}$
6. If x^4 occurs in the r^{th} term in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, then what is the value of r ?
(a) 4 (b) 8
(c) 9 (d) 10
7. After simplification, what is the number of terms in the expansion of $[(3x+y)^5]^4 - [3x-y]^4]^5$?
(a) 4 (b) 5
(c) 10 (d) 11
8. What is the coefficient of $x^3 y^4$ in $(2x + 3y^2)^5$?
(a) 240 (b) 360
(c) 720 (d) 1080
9. What is the last digit of $3^{3^{4n}} + 1$, where n is a natural number?
(a) 2 (b) 7
(c) 8 (d) None of these
10. If t_r is the r^{th} term in the expansion of $(1+x)^{101}$, then what is the ratio $\frac{t_{20}}{t_{19}}$ equal to
(a) $\frac{20x}{19}$ (b) $83x$
(c) $19x$ (d) $\frac{83x}{19}$
11. What is the value of ${}^8C_0 - {}^8C_1 + {}^8C_2 - {}^8C_3 + {}^8C_4 - {}^8C_5 + {}^8C_6 - {}^8C_7 + {}^8C_8$
(a) 0 (b) 1
(c) 2 (d) 2^8
12. What is the term independent of x in the expansion of $\left(1+x+2x^3\right)\left(\frac{3x^2}{3}-\frac{1}{3x}\right)^9$
(a) $1/3$
(b) $17/54$
(c) $1/4$
(d) No such term exists in the expansion
13. What is the coefficient of x^4 in the expansion of $(1+2x+3x^2+4x^3+\dots)^{1/2}$?
(a) $1/4$ (b) $1/16$
(c) 1 (d) $1/28$
14. Consider the following statements.
I. The coefficient of the middle term in the expansion of $(1+x)^8$ is equal to the middle term of $\left(x+\frac{1}{x}\right)^8$
II. The coefficient of the middle term in the expansion of $(1+x)^8$ is less than the coefficient of the fifth term in the expansion of $(1+x)^7$.
Which of the above statements is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
15. How many zeroes are there in $80!$
(a) 16 (b) 18
(c) 19 (d) 20
16. Find the remainder $\frac{16^{200}+5}{15}$
(a) 6 (b) 7
(c) 10 (d) 12
17. What is the sum of odd place coefficients in the binomial expansion of $(a+b)^n$
(a) 2^n (b) 2^{n+1}
(c) 2^{n-1} (d) 2
18. The coefficient of x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^8$ is
(a) ${}^8C_3 \frac{a^3}{b^5}$ (b) ${}^8C_3 \frac{a^5}{b^3}$
(c) ${}^8C_4 \frac{a^5}{b^3}$ (d) None of these
19. The coefficient of x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^8$
(a) $(-1)^5 {}^8C_5 \frac{a^3}{b^5}$ (b) $(-1)^5 {}^8C_5 \frac{a^5}{b^3}$
(c) ${}^8C_3 \frac{a^5}{b^3}$ (d) None of these
20. What is the coefficient of x^4 in the expansion of $\left(\frac{1-x}{1+x}\right)^2$?
(a) -16 (b) 16
(c) 8 (d) -8
21. What is the sum of the coefficient of all the terms in the expansion of $(45x-49)^4$?
(a) -256 (b) -100
(c) 100 (d) 256

22. What is the number of terms in the expansion of $(a + b + c)^n$, $n \in \mathbb{N}$?
 (a) $n + 1$ (b) $n + 2$ (c) $n(n + 1)$ (d) $\frac{(n+1)(n+2)}{2}$

ANSWER KEY

1.	d	2.	d	3.	d	4.	a	5.	d	6.	c	7.	c	8.	c	9.	d	10.	d
11.	a	12.	b	13.	c	14.	a	15.	c	16.	a	17.	c	18.	b	19.	a	20.	b
21.	d	22.	d																

Solutions

Sol.1. (d)

Sum of all odd position terms
 $= 2^{n-1} = 2^7 = 128$

Sol.2. (d)

Using binomial theorem $9^{200} = (1 + 8)^{2000}$

$$= 1 + 8.200 + \frac{200 \times 199}{2!} \times 8^2 + \dots$$

$$= 1 + 1600 + 1273600 + \dots$$

From above, it is clear that the last two digits of the number 9^{200} are 01

Sol.3. (d)

Using binomial theorem.

$$4^n - 3n = (1 + 3)n - 3n$$

$$1 + n.3 + \frac{n(n-1)}{2!} 3^2 + \dots - 3n$$

$$= 1 + \frac{n(n-1)}{2!} 3^2 + \frac{n(n-1)(n-2)}{3!} 3^3 + \dots$$

$$\Rightarrow 4^n - 3n = 9 \left\{ \frac{n(n-1)}{2!} + \dots \right\} + 1$$

Thus, when $4n - 3n$ is divided by 9, the remainder is 1.

Sol.4. (a)

$$1 - 2x + 3x^2 - 4x^3 + \dots$$

$$= (1 + x)^{-2}, \text{ so, } (1 - 2x + 3x^2 - 4x^3 + \dots \infty)^{-5}$$

$$= ((1+x)^{-2})^{-5} = (1+x)^{10} \Rightarrow T_{r+1} = {}^{10}C_r X^r$$

Putting $r = 5$, coefficient of

$$x^5 = {}^{10}C_5 = \frac{10!}{5!5!} = \frac{10!}{(5!)^2}$$

Sol.5. (d)

In the expansion of $\left(\frac{x\sqrt{y}}{3} - \frac{3}{y\sqrt{x}}\right)^{12}$, then

middle term is $\frac{12}{2} + 1 = 7^{\text{th}}$ term, $(r + 1)^{\text{th}}$

term,

$$T_{r+1} = {}^{12}C_r \left[\frac{x\sqrt{y}}{3}\right]^{12-r} \left[-\frac{3}{y\sqrt{x}}\right]^r$$

$$\therefore T_7 = T_{6+1} = {}^{12}C_6 \left(\frac{x\sqrt{y}}{3}\right)^6 \left(-\frac{3}{y\sqrt{x}}\right)^6$$

$$= {}^{12}C_6 \frac{x^6 y^3}{y^6 x^3} = {}^{12}C_6 x^3 y^{-3} = C(12, 6) x^3 y^{-3}$$

Sol.6. (c)

In the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, let T_r is the n^{th} term

$$T_r = {}^{15}C_{r-1} (x^4)^{15-r+1} \left(\frac{1}{x^3}\right)^{r-1}$$

$$= {}^{15}C_{r-1} x^{64-4-3r+3} = {}^{15}C_{r-1} x^{67-7r}$$

x^4 occurs in this term

$$\Rightarrow 4 = 67 - 7r$$

$$\Rightarrow 7r = 63$$

$$\Rightarrow r = 9$$

Sol.7. (c)

Given expression is:

$$[(3x + y)^5]^4 - [3x - y]^4]^5 = [(3x + y)^{20} - (3x - y)^{20}]$$

First and second expansion will have 21 terms each but odd terms in second expansion be $1^{\text{st}}, 3^{\text{rd}}, 5^{\text{th}}, \dots, 21^{\text{st}}$ will be equal and opposite to those of first expansion

Thus, the number of terms in the expansion of above expression is 10.

Sol.8. (c)

$$T_r = {}^nC_{r-1} (2x)^{r-1} (3y^2)^{n-r+1}$$

$$T_4 = {}^5C_3 (2x)^3 (3y^2)^2$$

$$= \frac{5!}{3!2!} 2^3 \cdot x^3 \cdot 9y^4 = \frac{5.4}{2.1} \times 8 \times 9 \times x^3 y^4 = 720 x^3 y^4$$

$$\therefore \text{coefficient } x^3 y^4 = 720$$

Sol.9. (d)

In 3^n , last digit is 3, if $n = 1$, 9 if $n = 2$, 7 if $n = 3$ and 1 if $n = 4$ and it is repeated after than

Given expression is $3^{3^{4n}} + 1$

$$\text{Let } x = 3^{3^{4n}} + 1 = 3^{81n} + 1$$

$$\Rightarrow x = 3^{80n} \cdot 3^{n+1}$$

Last digit of x will be decided by 3^n since 3^{80n} has power multiple of 4.

If $n = 1$ last digit is $3 + 1 = 4$

$n = 2$ last digit is $3^2 + 1 = 9 + 1 = 10$

So, last digit is zero

$n = 3$ last digit is $3^3 + 1 = 27 + 1 = 28$

last digit is 8.

If $n = 4$ last digit is $3^4 + 1 = 81 + 1 = 82$

last digit is 2.

So, there is no definite value of last digit.

Sol.10. (d)

We find r_n term:

t_r is the r^{th} term in the expansion of $(1 + x)^{101}$.

$$\therefore \frac{t_{20}}{t_{19}} = \frac{{}^{101}C_{19} \cdot x^{19}}{{}^{101}C_{18} \cdot x^{18}} = \frac{{}^{101}C_{19} x}{{}^{101}C_{18}}$$

$$= \frac{10!}{18!83!} x = \frac{83x}{19}$$

Sol.11. (a)

$$(1-x)^n = {}^nC_0 - {}^nC_1(x) + {}^nC_2 x^2 - {}^nC_3 x^3 + \dots + (-1)^n {}^nC_n$$

Put $x=1$ and $n=8$, then answer is 0

Sol.12. (b)

Given expansion

$$(1+x+2x^3) \left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$$

$$= (1+x+2x^3) \left[\left(\frac{3}{2}x^2\right)^9 - {}^9C_1 \left(\frac{3}{2}x^2\right)^8 \right.$$

$$\left. + {}^9C_6 \left(\frac{3}{2}x^2\right)^3 \left(\frac{1}{3x}\right)^6 - {}^9C_7 \left(\frac{3}{2}x^2\right)^2 \left(\frac{1}{3x}\right)^7 + \dots \right]$$

In the second bracket we have to search out terms of x^0 and $\frac{1}{x^3}$ which when multiplied

with the terms 1 and $2x^3$ in the first bracket will give a term independent of x . The term containing $\frac{1}{x}$ will not occur in the 2^{nd}

bracket.

$\therefore$ Term independent of x .

$$= 1 - \left[{}^9C_6 \frac{3^3}{2^3} \cdot \frac{1}{3^6} \right] - 2x^3 \left[{}^9C_7 \frac{3^2}{2^2} \cdot \frac{1}{3^7} \cdot \frac{1}{x^3} \right]$$

$$= \left[\frac{9.8.7}{1.2.3} \cdot \frac{1}{8.27} \right] - 2 \left[\frac{9.8}{1.2} \cdot \frac{1}{4.243} \right]$$

$$= \frac{7}{18} - \frac{2}{27} = \frac{17}{54}$$

Sol.13. (c)

Consider $(1 + 2x + 3x^2 + 4x^3 + \dots)^{1/2} = (1 - x^2)^{-1/2}$

As we know that $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$

$$\Rightarrow (1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$\therefore$ Required coefficient of x^4 is 1.

Sol.14. (a)

Statement I: Given expansion is $(1+x)^8$
Since, $n = 8$ is even

$\therefore \left(\frac{n}{2} + 1\right)$ th term is the middle term.

ie- $\left(\frac{8}{2} + 1\right)$ th = 5th term = middle term

Now, 5th term = ${}^8C_4 x^4 1^4 = {}^8C_4 x^4$

Coeff. of 5th term (middle term) = 8C_4

Now, consider the expansion $\left(x + \frac{1}{x}\right)^8$

It's middle term = 5th term

and 5th term = ${}^8C_4 x^4 \left(\frac{1}{x}\right)^4 = {}^8C_4$

Hence, statement I is correct

Statement II : Coeff. Of middle term in $(x+1)^8$ is ${}^8C_4 = \frac{8!}{4!4!} = 70$

Coeff of 5th term in $(1+x)^7 = {}^7C_4 = \frac{7!}{4!3!} = 35$

Sol.15. (c)

No of zeroes

$$= \left\lfloor \frac{80}{5} \right\rfloor + \left\lfloor \frac{80}{25} \right\rfloor + \left\lfloor \frac{80}{125} \right\rfloor$$

$$= 16 + 3 + 0 + \dots$$

$$= 19$$

Sol.16. (a)

$$= \frac{16^{200} + 5}{15} = \frac{(15+1)^{200} + 5}{15}$$

$$\text{Remainder} = 1+5=6$$

Sol.17. (c)

Sum of all coefficients = 2^n

Sum of odd place coefficients + sum of even place coefficient = 2^n and sum of odd place coefficients = sum of even place coefficients so sum of odd placed coefficients = $\frac{2^n}{2} = 2^{n-1}$

Sol.18. (b)

Let the term containing x^7 in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^8$ is T_{r+1}

$$\therefore T_{r+1} = {}^8C_r (ax^2)^{8-r} \left(\frac{1}{bx}\right)^r = {}^8C_r \frac{a^{8-r}}{b^r} x^{16-3r}$$

Since this term contains x^7 .

$$\therefore 16 - 3r = 7$$

$$\Rightarrow r = 3$$

$\therefore$ Coefficient of x^7 in the expansion of

$$\left(ax^2 + \frac{1}{bx}\right)^8 = {}^8C_3 \cdot a^5 / b^3$$

Sol.19. (a)

Also, the term containing x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^8$ is T_{R+1} .

$$T_{R+1} = {}^8C_R (ax)^{8-R} \left(-\frac{1}{bx^2}\right)^R$$

$$= (-1)^R {}^8C_R \frac{a^{8-R}}{b^R} x^{8-3R}$$

Since this term contains x^{-7} .

$$\therefore 8 - 3R = -7 \Rightarrow R = 5$$

$\therefore$ Coefficient of x^{-7} in the expansion of

$$\left(ax - \frac{1}{bx^2}\right)^8 = (-1)^5 {}^8C_5 \frac{a^3}{b^5}$$

Sol.20. (b)

$$\left(\frac{1-x}{1+x}\right)^2 = (1-x)^2 (1+x)^{-2}$$

$$= (1-2x+x^2)(1-2x+3x^2-4x^3+5x^4-\dots)$$

$$= (5+8+3)x^4 + \dots = 16x^4 + \dots$$

Sol.21. (d)

$$\text{Required sum} = (45-49)^4 = (-4)^4 = 256$$

Sol.22. (d)

$$\frac{(n+1)(n+2)}{2}$$

NDA PYQ

1. What is the middle term in the expansion of $\left(1 - \frac{x}{2}\right)^8$?
 (a) $\frac{35x^4}{8}$ (b) $\frac{17x^5}{8}$
 (c) $\frac{35x^5}{8}$ (d) None of these
 [NDA (I) - 2011]
 2. What is the ratio of coefficient of x^{15} to the term independent of x in $\left(x^2 + \frac{2}{x}\right)^{15}$?
 (a) $1/64$ (b) $1/32$
 (c) $1/16$ (d) $1/4$
 [NDA (II) - 2011]
 3. For all $n \in \mathbb{N}$, $2^{4n} - 15n - 1$ is divisible by:
 (a) 125 (b) 225
 (c) 450 (d) none of the above
 [NDA (II) - 2011]
 4. The value of the term independent of x in the expansion of $\left(x^2 - \frac{1}{x}\right)^9$ is
 (a) 9 (b) 18
 (c) 48 (d) 84
 [NDA (I) - 2012]
 5. In the expansion of $(1 + x)^n$, what is the sum of even binomial coefficient?
 (a) 2^n (b) 2^{n-1}
 (c) 2^{n+1} (d) none of these
 [NDA (I) - 2012]
 6. What is the sum of the coefficients in the expansion of $(1 + x)^n$?
 (a) 2^n (b) $2^n - 1$
 (c) $2^n + 1$ (d) $n + 1$
 [NDA (I) - 2013]
 7. How many terms are there in the expansion of $(1 + 2x + x^2)^{10}$?
 (a) 11 (b) 20
 (c) 21 (d) 30
 [NDA (II) - 2013]
 8. If n be a positive integer and $(1 + x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, then what is $a_0 + a_1 + a_2 + \dots + a_n$ equal to?
 (a) 1 (b) 2^n
 (c) 2^{n-1} (d) 2^{n+1}
 [NDA (II) - 2013]
- Directions (for next three):**
 In the expansion of $\left(x^3 - \frac{1}{x^2}\right)^n$, where n is a positive integer, the sum of the coefficients of x^5 and x^{10} is 0.
9. What is n equal to?
 (a) 5 (b) 10
 (c) 15 (d) None of these
 [NDA (I) - 2014]
 10. What is the value of the independent term?
 (a) 5005 (b) 7200
 (c) -5005 (d) -7200
 [NDA (I) - 2014]
 11. What is the sum of the coefficients of the two middle terms:

- (a) 0 (b) 1
 (c) -1 (d) None of these
 [NDA (I) - 2014]
- Directions (for next five):**
 Read the following information carefully and answer the questions given below.
 Consider the expansion $\left(x^2 + \frac{1}{x}\right)^{15}$.
12. What is the independent term in the given expansion?
 (a) 2103 (b) 3003
 (c) 4503 (d) None of these
 [NDA (II) - 2014]
 13. What is the ratio of coefficient of x^{15} to the term independent of x in the given expansion?
 (a) 1 (b) $1/2$
 (c) $2/3$ (d) $3/4$
 [NDA (II) - 2014]
 14. What is the sum of the coefficients of the middle terms in the given expansion?
 (a) C (15, 9) (b) C (16, 9)
 (c) C (16, 8) (d) None of these
 [NDA (II) - 2014]
 15. Consider the following statements
 I. There are 15 terms in the given expansion.
 II. The coefficient of x^{12} is equal to that of x^3 .
 Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
 [NDA (II) - 2014]
 16. Consider the following statements
 I. The term containing x^2 does not exist in the given expansion.
 II. The sum of the coefficients of all the terms in the given expansion is 2^{15} .
 Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) neither I nor II
 [NDA (II) - 2014]
 17. The coefficient of x^{99} in the expansion of $(x - 1)(x - 2)(x - 3) \dots (x - 100)$ is
 (a) 5050 (b) 5000
 (c) -5050 (d) -5000
 [NDA (II) - 2015]
 18. In the expansion of $\left(\sqrt{x} + \frac{1}{3x^2}\right)^{10}$ the value of constant term (independent of x) is
 (a) 5 (b) 8
 (c) 45 (d) 90
 [NDA (II) - 2015]
- Directions (for next three):**
 Read the following information carefully and answer the questions given below.
 Consider the expansion of $(1 + x)^{2n+1}$.
19. The sum of the coefficients of all the terms in the expansion is
 (a) 2^{2n-1} (b) 4^{n-1}
 (c) 2×4^n (d) None of these
 [NDA (II) - 2015]
 20. The average of the coefficients of the two middle terms in the expansion is



- (a) ${}^{2n+1}C_{n+2}$ (b) ${}^{2n+1}C_n$
(c) ${}^{2n+1}C_{n-1}$ (d) ${}^{2n}C_{n+1}$

[NDA (II) - 2015]

21. If the coefficient of x^r and x^{r+1} are equal in the expansion, then r is equal to

- (a) n (b) $\frac{2n-1}{2}$
(c) $\frac{2n+1}{2}$ (d) $n+1$

[NDA (II) - 2015]

22. The expansion of $(x-y)^n$, $n \geq 5$ is done in the descending powers of x . If the sum of the fifth and sixth terms is zero, then $\frac{x}{y}$ is equal to:

- (a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$
(c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$

[NDA (I) - 2017]

23. The number of terms in the expansion of $(x+a)^{100} + (x-a)^{100}$ after simplification is:

- (a) 202 (b) 101
(c) 51 (d) 50

[NDA (II) - 2017]

24. In the expansion of $(1+x)^{50}$, the sum of the coefficient of odd powers of x is:

- (a) 2^{26} (b) 2^{49}
(c) 2^{50} (d) 2^{51}

[NDA (II) - 2017]

25. If $n \in \mathbb{N}$, then $121^n - 25^n + 1900^n - (-4)^n$ is divisible by which one of the following:

- (a) 1904 (b) 2000
(c) 2002 (d) 2006

[NDA-2018(1)]

26. What is the number of non zero terms in the expansion of $(1+2\sqrt{3x})^{11} + (1-2\sqrt{3x})^{11}$?

- (a) 4 (b) 5
(c) 6 (d) 11

[NDA (I) - 2018]

27. If the coefficient of a^m and a^n in the expansion of $(1+a)^{m+n}$ are α and β then which one of the following is correct?

- (a) $\alpha = 2\beta$ (b) $\alpha = \beta$
(c) $2\alpha = \beta$ (d) $\alpha = (m+n)\beta$

[NDA (I) - 2018]

28. In the expansion of $(1+x)^{43}$, if the coefficient of $(2r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms are equal, then what is the value of r ($r \neq 1$)?

- (a) 5 (b) 14
(c) 21 (d) 22

[NDA (I) - 2018]

29. What is the greatest integer among the following by which the number $5^5 + 7^5$ is divisible?

- (a) 6 (b) 8
(c) 11 (d) 12

[NDA (I) - 2018]

30. Let the coefficient of the middle term of the binomial expansion of $(1+x)^{2n}$ be α and those of two middle terms of the binomial expansion of $(1+x)^{2n-1}$ be β and r which one of the following relation is correct?

- (a) $\alpha > \beta + r$ (b) $\alpha < \beta + r$
(c) $\alpha = \beta + r$ (d) $\alpha = \beta r$

[NDA (II) - 2018]

31. What is the coefficient of the middle term in the binomial expansion of $(2+3x)^4$:

- (a) 6 (b) 12
(c) 108 (d) 216

[NDA (II) - 2018]

32. Find the number of terms in the expansion of $[(2x-3y)^2(2x+3y)^2]^2$?

- (a) 4 (b) 5
(c) 8 (d) 16

[NDA (I) - 2019]

33. In the expansion of $(1+ax)^n$, first three terms are 1, $12x$ and $64x^2$, then $n = ?$

- (a) 6 (b) 9
(c) 10 (d) 12

[NDA (I) - 2019]

34. How many terms are there in the expression of $(1+2x+x^2)^5 + (1+4y+4y^2)^5$?

- (a) 12 (b) 20
(c) 21 (d) 22

[NDA (II) - 2019]

35. If the middle term in the expression of $\left(x^2 + \frac{1}{x}\right)^{2n}$ is $184756x^{10}$, then what is the value of n ?

- (a) 10 (b) 8
(c) 5 (d) 4

[NDA (II) - 2019]

36. If the constant term in the expansion of $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ is 405,

then what can be the value of k ?

- (a) ± 2 (b) ± 3
(c) ± 5 (d) ± 9

[NDA (II) - 2019]

37. What is the sum of last five coefficients in the expansion of $(1+x)^9$ when it is expanded in ascending power of x ?

- (a) 256 (b) 512
(c) 1024 (d) 2048

[NDA 2020]

38. The term independent of x in the binomial expansion of

$\left(\frac{2}{x^2} - \sqrt{x}\right)^{10}$ is equal to

- (a) 180 (b) 120
(c) 90 (d) 72

[NDA 2020]

39. If $(1+2x-x^2)^6 = a_0 + a_1x + a_2x^2 + \dots + a_{12}x^{12}$ then what is $a_0 - a_1 + a_2 - a_3 + \dots + a_{12}$ equal to

- (a) 32 (b) 64
(c) 2048 (d) 4096

[NDA 2020]

40. What is the coefficient of middle term in the expansion of $(1+4x+4x^2)^5$?

- (a) 8064 (b) 4032
(c) 2016 (d) 1008

[NDA (I) - 2021]

41. If $C_0, C_1, C_2, \dots, C_n$ are the coefficients in the expression of $(1+x)^n$, then what is the value of $C_1 + C_2 + C_3 + \dots + C_n$?

- (a) 2^n (b) $2^n - 1$
(c) 2^{n-1} (d) $2^n - 2$

[NDA (I) - 2021]

42. What is the sum of the coefficients of first and last terms in the expansion of $(1+x)^{2n}$, where n is a natural number?

- (a) 1 (b) 2

- (c) n (d) 2n

[NDA (I) - 2021]

43. How many terms are there in the expansion of $\left(\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2\right)^{21}$ where $a \neq 0, b \neq 0$?

- (a) 21 (b) 22
(c) 42 (d) 43

[NDA (II) - 2021]

44. Consider the expansion of $(1+x)^n$. Let p, q, r and s be the coefficient of first, second, nth and $(n+1)^{\text{th}}$ terms respectively. What is $(ps+qr)$ equal to?

- (a) $1+2n$ (b) $1+2n^2$
(c) $1+n^2$ (d) $1+4n$

[NDA (II) 2021]

45. How many terms are there in the expansion of $\left(1+\frac{2}{x}\right)^9 \times \left(1-\frac{2}{x}\right)^9$?

- (a) 9 (b) 10
(c) 19 (d) 20

[NDA (I) - 2022]

46. Consider the following statement in respect of the expansion of $(x+y)^{10}$:
1. Among all the coefficients of the terms, the coefficient of the 6th term has the highest value.
2. The coefficient of the 3rd term is equal to coefficient of the 9th term.

Which of the above statement is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

47. In the expansion of $\left(x+\frac{1}{x}\right)^{2n}$, what is the $(n+1)^{\text{th}}$ term from the end (when arranged in descending powers of x)?

- (a) $C(2n, n)x$ (b) $C(2n, n-1)x$
(c) $C(2n, n)$ (d) $C(2n, n-1)$

[NDA (I) - 2022]

48. What is $\sum_{r=0}^n 2^r C(n, r)$ equal to?

- (a) 2^n (b) 3^n
(c) 2^{2n} (d) 3^{2n}

[NDA 2022 (II)]

Consider the following for the next three (03) items that follow:

Consider the binomial expansion of $(p+qx)^9$:

49. What is the value of q if the coefficients of x^3 and x^6 are equal?

- (a) p (b) 9p
(c) $\frac{1}{p}$ (d) p^2

[NDA 2022 (II)]

50. What is the ratio of the coefficients of middle terms in the expansion (when expanded in ascending powers of x)?

- (a) pq (b) p/q
(c) $4p/5q$ (d) $1/(pq)$

[NDA 2022 (II)]

51. Under what condition of the coefficients of x^2 and x^4 are equal?

- (a) $p:q = 7:2$ (b) $p^2:q^2 = 7:2$
(c) $p:q = 2:7$ (d) $p^2:q^2 = 2:7$

[NDA 2022 (II)]

52. How many terms are there in the expansion of $(3x-y)^4$ $(x+3y)^4$?

- (a) 9 (b) 12
(c) 15 (d) 17

[NDA - 2023 (1)]

53. If the sixth term in the binomial expansion of $\left(x^{-\frac{8}{3}} + x^2 \log_{10} x\right)^8$ is 5600, then what is the value of x?

- (a) 6 (b) 8
(c) 9 (d) 10

[NDA - 2023 (1)]

Consider the following for the next (03) items that follow:

Let $(1+x)^n = 1 + T_1x + T_2x^2 + T_3x^3 + \dots + T_nx^n$.

54. What is $T_1 + 2T_2 + 3T_3 + \dots + nT_n$ equal to?

- (a) 0 (b) 1
(c) 2^n (d) $n2^{n-1}$

[NDA-2023 (2)]

55. What is $1 - T_1 + 2T_2 - 3T_3 + \dots + (-1)^n nT_n$ equal to?

- (a) 0 (b) -2^{n-1}
(c) $n2^{n-1}$ (d) 1

[NDA-2023 (2)]

56. What is $T_1 + T_2 + T_3 + \dots + T_n$ equal to?

- (a) 2^n (b) $2^n - 1$
(c) 2^{n-1} (d) $2^n + 1$

[NDA-2023 (2)]

57. What is the coefficient of x^{10} in the expansion of $(1-x^2)^{20} \left(2-x^2-\frac{1}{x^2}\right)^{-5}$?

- (a) -1
(b) 1
(c) 10
(d) Coefficient of x^{10} does not exist

[NDA-2024 (1)]

58. What is the remainder when 2^{120} is divided by 7?

- (a) 1 (b) 3
(c) 5 (d) 6

[NDA-2024 (1)]

59. If the 4th term in the expansion of $\left(mx + \frac{1}{x}\right)^n$ is $\frac{5}{2}$, then what is the value of mn?

- (a) -3 (b) 3
(c) 6 (d) 12

[NDA-2024 (1)]

60. In a binomial expansion of $(x+y)^{2n+1} (x-y)^{2n+1}$, the sum of middle terms is zero. What is the value of $\left(\frac{x^2}{y^2}\right)$?

- (a) 1 (b) 2
(c) 4 (d) 8

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

Let $(8+3\sqrt{7})^{20} = U + V$ and $(8-3\sqrt{7})^{20} = W$, where U is an integer and $0 < V < 1$.

61. What is $V + W$ equal to ?

- (a) 8 (b) 4
(c) 2 (d) 1

[NDA-2024 (2)]

62. What is the value of $(U+V)W$?

- (a) $1/2$ (b) 1
(c) $3/2$ (d) 2

[NDA-2024 (2)]



- 63.** What is the remainder when $7^n - 6n$ is divided by 36 for $n = 100$?
(a) 0 (b) 1 (c) 2 (d) 6
[NDA-2024 (2)]
- 64.** In the expansion of $(1+x)^p(1+x)^q$, if the coefficient of x^3 is 35, then what is the value of $(p+q)$?
(a) 5 (b) 6 (c) 7 (d) 8
[NDA-2024 (2)]
- 65.** What is the number of rational terms in the expansion of $\left(3^{\frac{1}{2}} + 5^{\frac{1}{4}}\right)^{12}$?
(a) 2 (b) 4 (c) 6 (d) 12
[NDA-2025 (1)]
- 66.** If the sum of binomial coefficients in the expansion of $(x+y)^n$ is 256, then the greatest binomial coefficient occurs in which one of the following terms?
(a) Third (b) Forth (c) Fifth (d) Ninth
[NDA-2025 (1)]
- 67.** If $k < (\sqrt{2} + 1)^3 < k + 2$, where k is a natural number, then what is the value of k ?
(a) 11 (b) 13 (c) 15 (d) 17
[NDA-2025 (1)]
- 68.** What is the remainder when 5^{99} is divided by 13?
(a) 10 (b) 9 (c) 8 (d) 6
[NDA-2025 (2)]
- 69.** Which one of the following is the greatest coefficient in the expansion of $((1+x)^{100})$?
(a) The coefficient of x^{100} (b) The coefficient of x^{99} (c) The coefficient of x^{51} (d) The coefficient of x^{50}
[NDA-2025 (2)]

ANSWER KEY

1.	a	2.	b	3.	b	4.	d	5.	b	6.	a	7.	c	8.	b	9.	c	10.	c
11.	a	12.	b	13.	a	14.	c	15.	b	16.	c	17.	c	18.	a	19.	c	20.	b
21.	a	22.	b	23.	c	24.	b	25.	b	26.	c	27.	b	28.	b	29.	d	30.	c
31.	d	32.	b	33.	b	34.	c	35.	a	36.	b	37.	a	38.	a	39.	b	40.	a
41.	b	42.	b	43.	d	44.	c	45.	b	46.	c	47.	c	48.	b	49.	a	50.	b
51.	b	52.	a	53.	d	54.	d	55.	d	56.	b	57.	a	58.	a	59.	b	60.	a
61.	d	62.	b	63.	b	64.	c	65.	b	66.	c	67.	b	68.	c	69.	d		

Solutions

Sol.1. (a)

Given expression = $\left(1 - \frac{x}{2}\right)^8$

Here, $n = 8$, which is an even number

So, $\left(\frac{8}{2} + 1\right)$ the term is the middle term.

Now, $T_5 = {}^8C_4(1)^{8-4}\left(-\frac{x}{2}\right)^4$

$= \frac{8.7.6.5}{4.3.2.1} \left(\frac{x^4}{16}\right) = \frac{35x^2}{8}$

Sol.2. (b)

Binomial expression = $\left(x^2 + \frac{2}{x}\right)^{15}$

General term,

$T_{r+1} = {}^{15}C_r (x^2)^{15-r} \left(\frac{2}{x}\right)^r$

$= {}^{15}C_r x^{30-2r} (2)^r \cdot x^{-r} = {}^{15}C_r 2^r x^{30-3r}$

...(i)

Put $30 - 3r = 15 \Rightarrow 3r = 15 \Rightarrow r = 5$

$\therefore T_{5+1} = {}^{15}C_5 \cdot 2^5 \cdot X^{30-30 \times 5} = {}^{15}C_5 2^5 \cdot X^{15}$

Coefficient of x^{15} in $\left(x^2 + \frac{2}{x}\right)^{15} = {}^{15}C_r \cdot 2^5$

...(ii)

For coefficient of x^0 ,

Put, $30 - 3r = 0 \Rightarrow r = 10$

$\therefore T_{10+1} = {}^{15}C_{10} 2^{10} \cdot X^{30-3 \times 10} = {}^{15}C_{10} 2^{10} \cdot x^0$...(iii)

$\therefore$ Ratio of coefficient of x^{15} to the term independent of x

$= \frac{{}^{15}C_5 \cdot 2^5}{{}^{15}C_{10} 2^{10}} = \frac{{}^{15}C_{10} \cdot 2^5}{{}^{15}C_{10} \cdot 2^{10}} = \frac{1}{2^5} = \frac{1}{32}$

$(\because {}^nC_r = {}^nC_{n-r})$

Sol.3. (b)

$2^{4n} - 15n - 1 = 16^n - 15n - 1$

$= (15 + 1)^n - 15n - 1$

$= [{}^nC_0 15^n + {}^nC_1 15^{n-1} + \dots + {}^nC_{n-2} 15^2 + {}^nC_{n-1} 15 + {}^nC_n] - 15n - 1$

$= [{}^nC_0 15^n + {}^nC_1 15^{n-1} + \dots + {}^nC_{n-2} 15^2]$

15^2 common so 225 will always divide the expression.

Sol.4. (d)

Given, $\left(x^2 - \frac{1}{x}\right)^9$

General term, $T_{r+1} = {}^nC_r (1)^{n-r} x^r$, in $(1+x)^n$

Similarly,

$T_{r+1} = {}^9C_r (x^2)^{9-r} \cdot \left(-\frac{1}{x}\right)^r$, in $\left(x^2 - \frac{1}{x}\right)^9$

$= {}^9C_r x^{18-2r} (-1)^r x^{-r}$

$= {}^9C_r x^{(18-3r)} x^{-r}$

...(i)

For independent term,

Put $18 - 3r = 0 \Rightarrow 3r = 18 \Rightarrow r = 6$

$\therefore T_{(6+1)} = {}^9C_6 x^{(18-18)} \cdot (-1)^6$

$T_7 = {}^9C_6 \cdot 1 = \frac{9.8.7}{3.2.1} = 84$

Sol.5. (b)

sum of even or odd coefficient = 2^{n-1}

Sol.6. (a)

For the sum of the coefficients in the expansion of $(1+x)^n$

Put $x = 1$

$(1+x)^n = (1+1)^n = 2^n$

Which is the required sum of the coefficient.

Sol.7. (c)

Given, $(1+2x+x^2)^{10} = \{(1+x)^2\}^{10} = (1+x)^{20}$

$\therefore$ total terms = $20 + 1 = 21$

Sol.8. (b)

Given that,

$(1+x)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$

Put $x = 1$

$(1+1)^n = a_0 + a_1 + a_2 + \dots + a_n$

$$\Rightarrow a_0 + a_1 + a_2 + \dots + a_n = 2^n$$

Sol.9. (c)

$${}^nC_{\left(\frac{3n}{2}-2\right)} = {}^nC_{\left(\frac{3n}{5}-1\right)} \quad [\text{from Equation (ii)}]$$

$$\Rightarrow n\left(\frac{3n}{5}-2\right) + \left(\frac{3n}{5}-1\right)$$

$$(\because {}^nC_x = {}^nC_y \Rightarrow n = x + y)$$

$$\Rightarrow n = \frac{6n}{5} - 3$$

$$\Rightarrow \frac{6n}{5} - n = 3 \Rightarrow \frac{n}{5} = 3$$

$$\therefore n = 15$$

Sol.10. (c)

For the independent term,

$$\text{Put } 3n - 5r = 0 \quad [\text{from eq. (i)}]$$

$$\Rightarrow 5r = 3n = 3 \times 15 \quad (\because n = 15)$$

$$\Rightarrow 5r = 3 \times 3 \times 5$$

$$\therefore r = 9$$

Now, put the value of r in Eq. (i), we get,

$$T_{9+1} = {}^{15}C_9 (-1)^9 \cdot x^{(3 \times 15 - 5 \times 9)}$$

$$\Rightarrow T_{10} = - {}^{15}C_9 \cdot x^0 = - {}^{15}C_9$$

$$\Rightarrow T_{10} = - {}^{15}C_6 \quad (\because {}^nC_r = {}^nC_{n-r})$$

$$= - \frac{15!}{6!9!} \left(\because \frac{n!}{r!(n-r)!} \right)$$

$$= - \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= - 5005$$

Sol.11. (a)

Since, $n = 15$

$$\therefore \text{Total term in the expansion of } \left(x^3 - \frac{1}{x^2} \right)^{15} \text{ is}$$

$$16$$

$$\text{So, middle term} = \left(\frac{n}{2} \right) \text{th term and } \left(\frac{n}{2} + 1 \right) \text{th}$$

term

$$= \left(\frac{16}{2} \right) \text{th term and } \left(\frac{16}{2} + 1 \right) \text{th term}$$

$$= 8^{\text{th}} \text{ term and } 9^{\text{th}} \text{ term}$$

$$\text{So, } \left(x^3 - \frac{1}{x^2} \right)^{15} \text{ has two middle terms}$$

$$T_8 = T_{(7+1)} = {}^{15}C_8 \cdot (-1)^7 \cdot x^{(3 \times 15 - 5 \times 7)}$$

$$= - {}^{15}C_7 \cdot x^{10} \quad [\text{from eq. (i)}]$$

$$\text{and } T_9 = T_{(8+1)} = {}^{15}C_8 \cdot (-1)^8 \cdot x^{(3 \times 15 - 5 \times 8)}$$

$$= {}^{15}C_8 \cdot x^5 \quad [\text{from eq. (ii)}]$$

Now, the sum of the coefficients of the two middle terms

$$= - {}^{15}C_7 + {}^{15}C_8$$

$$= - {}^{15}C_7 + {}^{15}C_7 \quad (\because {}^nC_r = {}^nC_{n-r})$$

$$= 0$$

Sol.12. (b)

$$\text{We have, } \left(x^2 + \frac{1}{x} \right)^{15}$$

$$T_{r+1} = {}^{15}C_r (x^2)^{15-r} \left(\frac{1}{x} \right)^r$$

$$= {}^{15}C_r x^{30-2r-r} = {}^{15}C_r x^{30-3r}$$

For independent term,

$$30 - 3r = 0 \Rightarrow r = 100$$

Put $r = 10$, we get

$$T_{10+1} = {}^{15}C_{10} = \frac{15!}{10!5!}$$

$$= 15 \times 14 \times 13 \times 12 \times 11 \times 10! = 3003$$

Sol.13. (a)

For coefficient of x^{15}

$$30 - 3r = 15 \Rightarrow r = 5$$

So, the coefficient of x^{15} is ${}^{15}C_5$.

and coefficient of independent of x is

$$30 - 3r = 0 \Rightarrow r = 10$$

So, coefficient of independent of x is ${}^{15}C_{10}$.

$$\therefore \text{Required ratio} = \frac{{}^{15}C_5}{{}^{15}C_{10}}$$

$$= \frac{{}^{15}C_5}{{}^{15}C_5} \quad (\because {}^nC_r = {}^nC_{n-r}) = 1$$

Sol.14. (c)

$$\text{We have, } \left(x^2 + \frac{1}{x} \right)^{15}$$

Since, n is odd

So, it has two middle terms T_8 and T_9

$$\therefore T_8 + T_9 = {}^{15}C_7 + {}^{15}C_8$$

$$= {}^{16}C_8 \quad (\because {}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r)$$

Sol.15. (b)

1. We know that, $(a+b)^n$ have total number of terms is $n + 1$.

$$\text{So, } \left(x^2 + \frac{1}{x} \right)^{15} \text{ have 16 terms}$$

Hence, Statement 1 is not correct.

2. For coefficient of x^{12} ,

$$30 - 3r = 12$$

$$\Rightarrow r = 6 \Rightarrow {}^{15}C_6$$

and for coefficient of x^3 ,

$$30 - 3r = 3 \Rightarrow r = 9 \Rightarrow {}^{15}C_9$$

$$\therefore {}^{15}C_6 = {}^{15}C_9$$

Hence, statement 2 is correct.

Sol.16. (c)

1. For coefficient of x^2

$$30 - 3r = 2 \Rightarrow r = \frac{28}{3}, r \notin \mathbb{N}$$

So, x^2 does not exist in the expansion.

Hence, statement 1 is correct

2. Now,

$$\left(x^2 + \frac{1}{4} \right)^{15} = {}^{15}C_0 (x^2)^{15} + {}^{15}C_1 (x^2)^{14} \left(\frac{1}{x} \right) + \dots + {}^{15}C_{15} \left(\frac{1}{x} \right)^{15}$$

Put $x = 1$ both sides, we get

$$(1+1)^{15} = {}^{15}C_0 + {}^{15}C_1 + \dots + {}^{15}C_{15}$$

$$\Rightarrow 2^{15} = {}^{15}C_0 + {}^{15}C_1 + \dots + {}^{15}C_{15}$$

Hence, statement 2 is correct.

Sol.17. (c)

$$(x-1)(x-2)(x-3) \dots (x-100)$$

Then the coefficient of x^{99} is given as

$$-1-2-3-4 \dots -100$$

$$= -1(1+2+3+ \dots +100)$$

$$= - \left(100 \times \frac{101}{2} \right)$$

$$= (-5050)$$

Sol.18. (a)

Independent term in the expansion of

$$\left(\sqrt{x} + \frac{1}{3x^2} \right)^{10}$$

$$T_{r+1} = {}^{10}C_r x^{1/2(10-r)} \times \left(\frac{1}{3x^2} \right)^r$$

$$= {}^{10}C_r \cdot x^{1/2(10-r) - 2r} \left(\frac{1}{3} \right)^r \quad \dots(i)$$

Term independent from x

$$= \frac{1}{2} (10-r) - 2r = 0$$

$$\Rightarrow 10-r-4r = 0$$

$$\Rightarrow r = 2$$

Putting the value of r in eq. (ii)

$$T_{2+1} = {}^{10}C_2 \times \left(\frac{1}{3} \right)^2 = \frac{10 \times 9 \times 8!}{2 \times 8!} \times \frac{1}{9} = 5$$

Sol.19. (c)

The sum of the coefficient $= 2^{2n+1} = (2^2)^n \times 2 = 2 \times 4^n$.

Sol.20. (b)

Here total number of terms $= 2n + 1 + 1 = 2n + 2$

$\therefore$ middle terms are $\left(\frac{2n+2}{2} \right)$ th term and

$$\left(\frac{2n+4}{2} \right) \text{th term}$$

i.e., $(n+1)$ th and $(n+2)$ th term

$$\text{Now, } T_{n+1} = {}^{2n+1}C_n \cdot x^n$$

$$\text{and } T_{n+2} = {}^{2n+1}C_{n+1} \cdot x^{n+1}$$

$$\therefore \text{required average} = \frac{{}^{2n+1}C_n + {}^{2n+1}C_{n+1}}{2}$$

$$= \frac{{}^{2n+1}C_n + {}^{2n+1}C_{n+1}}{2}$$

{ $\because$ these two coefficients are equal. }

$$= {}^{2n+1}C_n$$

Sol.21. (a)

$$(1+x)^{2n+1}$$

Given the coefficient of x^r and x^{r+1} are equal

$$T_{r+1} = {}^{2n+1}C_r \cdot (1)^{2n+1-r} \cdot (x)^r$$

$$= {}^{2n+1}C_r \cdot (1) \cdot x^r \quad \dots(i)$$

The coefficient of $x^r = {}^{2n+1}C_r$

and the coefficient of x^{r+1}

$$= {}^{2n+1}C_{r+1} \cdot (1)^{2n+1-r-1}$$

$$= {}^{2n+1}C_{r+1} \quad \dots(ii)$$

According to the question,

the coefficient of x^r and x^{r+1} are equal

$$\Rightarrow {}^{2n+1}C_r = {}^{2n+1}C_{r+1}$$

$$\Rightarrow \frac{(2n+1)!}{2n+1-r!} = \frac{(2n+1)!}{(2n-r)!(r+1)!}$$

$$\Rightarrow (2n-r)!(r+1)! = (2n+1-r)!(r)!$$

$$\Rightarrow (2n-r)!(r+1)r! = (2n+1-r)(2n-r)!r!$$

$$\Rightarrow r+1 = 2n+1-r$$

$$\Rightarrow 2r = 2n$$

$$\Rightarrow r = n$$

Sol.22. (b)

In the expansion of $(x-y)^n$, in decreasing power of x,

$$T_5 = T_{4+1} = {}^nC_4 \cdot x^{n-4} \cdot (-y)^5$$

$$\text{and } T_6 = T_{5+1} = {}^nC_5 \cdot x^{n-5} \cdot (-y)^5$$

According to the question,

$$T_5 + T_6 = 0$$

$$\Rightarrow {}^nC_4 \cdot n^{n-4} \cdot y^4 + {}^nC_5 \cdot x^{n-5} \cdot (-y)^5 = 0$$

$$\Rightarrow {}^nC_4 \cdot n^{n-4} \cdot y^4 = {}^nC_5 \cdot x^{n-5} \cdot y^5$$

$$\Rightarrow \frac{x^{n-4}}{x^{n-5}} \cdot \frac{y^4}{y^5} = \frac{{}^nC_5}{{}^nC_4}$$

$$\Rightarrow \frac{x}{y} = \frac{n!}{(n-5)! \cdot 5!} \times \frac{(n-4)! \cdot 4!}{n!} = \frac{n-4}{5}$$

Sol.23. (c)

The number of terms in the expansion of $(x+a)^n + (x-a)^n$. If n is even equal to

$$\left(\frac{n}{2} + 1 \right) \text{ i.e., } \left(\frac{100}{2} + 1 \right) = 51$$

Sol.24. (b)

In $(1+x)^{50}$ the coefficient of odd powers =
 $\frac{2^{50}}{2} = 2^{49}$

Sol.25. (b)

$$121^n - 25^n + 1900^n - (-4)^n$$

Put $n = 1$, we will get 2000

Which is divisible by 2000 according to second option

Sol.26. (c)

$$(a+b)^{11} + (a-b)^{11}$$

$\Rightarrow (6 \text{ even terms} + 6 \text{ even term}) + (6 \text{ even terms} - 6 \text{ odd terms})$

$\Rightarrow 6 \text{ terms}$

Sol.27. (b)

$$(1+a)^m (1+a)^n$$

$$({}^m C_0 + {}^m C_1 a + {}^m C_2 a^2 + \dots + {}^m C_m a^m) ({}^n C_0 + {}^n C_1 a + \dots + {}^n C_n a^n)$$

$$\text{Coefficient of } a^m = {}^m C_m \cdot {}^n C_0 = 1 \Rightarrow \alpha$$

$$\text{Coefficient of } a^n = {}^n C_n \cdot {}^m C_0 = 1 \Rightarrow \beta$$

Sol.28. (b)

$$(2r+1) + (r+2) = 43 + 2, \quad 3r+3 = 45$$

$$r = \frac{42}{3} = 14$$

Sol.29. (d)

$(a^n + b^n)$ is always divisible by $a + b$ when n is odd.

Sol.30. (c)

$$\text{Coefficient of middle term of } (1+x)^{2n} = {}^{2n} C_n = \alpha$$

$$\text{Coefficient of middle term of } (1+x)^{2n-1} = {}^{2n-1} C_{n-1} = \beta \text{ and } {}^{2n-1} C_n = r$$

$$\therefore {}^{2n-1} C_{n-1} + {}^{2n-1} C_n = {}^{2n} C_n$$

$$\beta + r = \alpha$$

Sol.31. (d)

Middle term is 3^{rd} term

$$T_3 = {}^4 C_2 (2)^2 (3x)^2 = \frac{4 \times 3}{2} \times 4 \times 9x^2$$

$$= 216x^2$$

Coefficient = 216

Sol.32. (b)

$$[(2x-3y)^2(2x+3y)^2] = (4x^2-9y^2)^4$$

no. of terms in expansion = 5

Sol.33. (b)

In the expansion of $(1+ax)^n$, first three terms are 1, $12x$ and $64x^2$, then $n = ?$

$$(1+ax)^n = 1 + nax + \frac{n(n-1)}{2} a^2 x^2 + \dots$$

$$\Rightarrow an = 12 \text{ and } \frac{n(n-1)}{2} a^2 = 64$$

by solving these equations $n = 9$

Sol.34. (c)

$$(1+x)^{10} + (1+2y)^{10}$$

11 terms in each expression but 1 term constant is common so there is 21 terms total.

Sol.35. (a)

Middle term is $(n+1)^{\text{th}}$ term

$$T_{n+1} = {}^{2n} C_n (x^2)^n (1/x)^n$$

$$= {}^{2n} C_n (x)^n = 184756x^{10}$$

$n = 10$ by comparing the powers of x .

Sol.36. (b)

General term, $T_{r+1} = {}^n C_r (a)^{n-r} (b)^r$, in $(a+b)^n$

$$T_{r+1} = {}^{10} C_r (\sqrt{x})^{10-r} \cdot \left(-\frac{k}{x^2}\right)^r,$$

$$= {}^{10} C_r x^{5-r/2} (-k)^r x^{-2r} \dots (i)$$

For independent term,

$$\text{Put } 5 - r/2 - 2r = 0 \Rightarrow r = 3$$

$$T_{2+1} = {}^{10} C_2 (\sqrt{x})^{10-2} \cdot \left(-\frac{k}{x^2}\right)^2,$$

$$45 k^2 = 405$$

$$k = \pm 3$$

Sol.37. (a)

sum of coefficients of all terms = $2^9 = 512$

there is total 10 terms and sum of first five terms

= sum of last five terms

so sum of last five terms = 256

Sol.38. (a)

$$\left(\frac{2}{x^2} - \sqrt{x}\right)^{10}$$

General term, $T_{r+1} = {}^n C_r (a)^{n-r} (b)^r$, in $(a+b)^n$

$$T_{r+1} = {}^{10} C_r \left(\frac{2}{x^2}\right)^{10-r} \cdot (\sqrt{x})^r$$

$$= {}^{10} C_r \cdot 2^{10-r} x^{20-2r} (x)^{r/2} \dots (i)$$

For independent term,

$$\text{Put } 20 - 5r/2 = 0 \Rightarrow r = 8$$

$$T_{8+1} = {}^{10} C_8 \left(\frac{2}{x^2}\right)^{10-8} \cdot (\sqrt{x})^8 = 45.2^2$$

$$= 180$$

Sol.39. (b)

$$\text{If } (1 + 2x - x^2)^6 = a_0 + a_1 x + a_2 x^2 + \dots + a_{12} x^{12}$$

put $x = -1$ both sides

$$a_0 - a_1 + a_2 - a_3 + \dots + a_{12} = (1-2-1)^6 = 64$$

Sol.40. (a)

$$(1+4x+4x^2)^5 = (1+2x)^{10}$$

There will be 11 terms in expansion, and 6^{th} term will be its middle term

$$\Rightarrow {}^{10} C_5 (1)^5 (2x)^5 = 8064x^5$$

Coefficient of middle term is 8064

Sol.41. (b)

$${}^n C_0 + {}^n C_1 + {}^n C_2 + \dots + {}^n C_n = 2^n$$

$${}^n C_1 + {}^n C_2 + \dots + {}^n C_n = 2^n - 1$$

Sol.42. (b)

$$(1+x)^{2n}$$

coeff. of first and last terms is 1 and 1

so sum is 2

Sol.43. (d)

$$\left(\frac{a^2}{b^2} + \frac{b^2}{a^2} + 2\right)^{21} = \left[\left(\frac{a}{b} + \frac{b}{a}\right)^2\right]^{21} = \left(\frac{a}{b} + \frac{b}{a}\right)^{42}$$

Numbers of terms = 43

Sol.44. (c)

$$(1+x)^n$$

$$1^{\text{st}} \text{ term} = 1$$

$$2^{\text{nd}} \text{ term} = {}^n C_1 x = nx$$

$$n^{\text{th}} \text{ term} = {}^n C_{n-1} x^{n-1} = nx^{n-1}$$

$$(n+1)^{\text{th}} \text{ term} = {}^n C_n x^n = x^n$$

$$p=1, q=n, r=n, S=1$$

$$PS + qr = 1 + n^2$$

Sol.45. (b)

$$\left(1 + \frac{2}{n}\right)^9 \left(1 - \frac{2}{x}\right)^9 = \left(1 - \frac{4}{x^2}\right)^9$$

total 10 terms in expansion.

Sol.46. (c)

1. middle term of $(x+y)^{10}$ is 6^{th} term and we know that coefficient of middle term is largest.

2. coefficient of 3^{rd} term = ${}^{10} C_2$ coefficient of 9^{th}

$$\text{term} - {}^{10} C_8 \quad {}^{10} C_2 = {}^{10} C_8$$

Both statements are correct.

Sol.47. (c)

$$(n+1)^{\text{th}} \text{ term of } \left(n + \frac{1}{x}\right)^{2n}$$

$$= {}^{2n} C_n x^n \left(\frac{1}{x}\right)^n$$

$$= {}^{2n} C_n$$

Sol.48. (b)

$$\sum_{r=0}^n 2^r C(n, r) \cdot 2^n$$

$$= {}^n C_0 2^0 + {}^n C_1 2^1 + {}^n C_2 2^2 + \dots + {}^n C_n$$

$$(a+b)^n = {}^n C_0 a^n + {}^n C_1 a^{n-1} b + {}^n C_2 a^{n-2} b^2 + \dots + {}^n C_n b^n$$

$$+ \dots + {}^n C_n b^n$$

Put $a = 1$ $b = 2$

$$\Rightarrow (1+2)^n = {}^n C_0 2^0 + {}^n C_1 2^1 + {}^n C_2 2^2 + \dots + {}^n C_n 2^n$$

$$\Rightarrow 3^n = {}^n C_0 2^0 + {}^n C_1 2^1 + {}^n C_2 2^2 + \dots + {}^n C_n 2^n$$

Sol.49. (a)

General term $(r+1)^{\text{th}}$ term is

$$\Rightarrow {}^n C_r p^{n-r} \cdot q^r x^r$$

For Coefficient of x^3 , $r = 3$

$$\text{Then coeff of } x^3 \text{ is } \Rightarrow {}^9 C_3 p^{9-3} \cdot q^3$$

For Coefficient of x^6 , $r = 6$

$$\text{Then coeff of } x^6 \text{ is } \Rightarrow {}^9 C_6 p^{9-6} \cdot q^6$$

coefficients of x^3 and x^6 are equal

$$\Rightarrow {}^9 C_3 \cdot p^6 \cdot q^3 = \Rightarrow {}^9 C_6 \cdot p^3 \cdot q^6$$

$$\Rightarrow p^3 = q^3$$

$$\Rightarrow p = q$$

Sol.50. (b)

Middle term exit at 5^{th} and 6^{th} position

5^{th} term

$$\Rightarrow {}^9 C_4 p^{9-4} \cdot q^4 = {}^9 C_4 p^5 \cdot q^4$$

6^{th} term

$$\Rightarrow {}^9 C_5 p^{9-5} \cdot q^5 = {}^9 C_5 p^4 \cdot q^5$$

Ratio of above two expressions is

$$\Rightarrow \frac{{}^9 C_4 p^5 \cdot q^4}{{}^9 C_5 p^4 \cdot q^5} = \frac{p}{q}$$

Sol.51. (b)

$$\text{General term } (r+1)^{\text{th}} \text{ term is } \Rightarrow {}^n C_r p^{n-r} \cdot q^r x^r$$

For Coefficient of x^2 , $r = 2$

$$\text{Then coeff of } x^2 \text{ is } \Rightarrow {}^9 C_2 p^{9-2} \cdot q^2 = {}^9 C_2 p^7 \cdot q^2$$

For Coefficient of x^4 , $r = 4$

$$\text{Then coeff of } x^4 \text{ is } \Rightarrow {}^9 C_4 p^{9-4} \cdot q^4 = {}^9 C_4 p^5 \cdot q^4$$

coefficients of x^2 and x^4 are equal

$$\Rightarrow {}^9 C_2 p^7 \cdot q^2 = {}^9 C_4 p^5 \cdot q^4$$

$$\Rightarrow 36p^2 = 126q^2$$

$$\Rightarrow 2p^2 = 7q^2$$

$$\Rightarrow p^2 : q^2 = 7 : 2$$

Consider the binomial expansion of $(p+qx)^9$

Sol.52. (a)

Maximum power of x in the expansion of $(3x-y)^4 (x+3y)^4$ is x^8 and minimum power of x will be x^0

When we write all these terms in descending power of x , we will get 9 terms.

Sol.53. (d)

Given

$$\left(x^{-\frac{8}{3}} + x^2 \log_{10} x\right)^8$$

6^{th} term of above binomial expansion $(a+b)^n$ is

$$\Rightarrow {}^8C_5 \cdot x^{-\frac{8}{3}} \cdot (x^2 \log_{10} x)^5$$

$$\Rightarrow 56 \left(x^{-8} \right) x^{10} (\log_{10} x)^5 = 5600$$

$$\Rightarrow x^2 (\log_{10} x)^5 = 100$$

Now by checking options we can say $x = 10$

Sol.54. (d)

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n b^n$$

$$+ \dots + {}^nC_n b^n$$

$$(a+b)^n = T_0 a^n + T_1 a^{n-1} b + T_2 a^{n-2} b^2 + \dots$$

$$+ \dots + T_n b^n$$

$$(1+x)^n = T_0 + T_1 x + T_2 x^2 + T_3 x^3 + \dots + T_n x^n$$

Differentiate both side w.r.t. x

$$n(1+x)^{n-1} = T_1 + 2T_2 x + 3T_3 x^2 + \dots + nT_n x^{n-1} \quad \dots \dots \dots (i)$$

Put $x = 1$

$$n(1+1)^{n-1} = T_1 + 2T_2 + 3T_3 + \dots + nT_n$$

$$n2^{n-1} = T_1 + 2T_2 + 3T_3 + \dots + nT_n$$

Sol.55. (d)

From equation (i) in above question

$$n(1+x)^{n-1} = T_1 + 2T_2 x + 3T_3 x^2 + \dots + nT_n x^{n-1}$$

Put $x = -1$

$$0 = T_1 - 2T_2 + 3T_3 - \dots + (-1)^n nT_n$$

Multiply both side with -1

$$0 = -T_1 + 2T_2 - 3T_3 + \dots + (-1)^n nT_n$$

Add 1 both side

$$1 = 1 - T_1 + 2T_2 - 3T_3 + \dots + (-1)^n nT_n$$

Sol.56. (b)

$$(1+x)^n = T_0 + T_1 x + T_2 x^2 + T_3 x^3 + \dots + T_n x^n$$

put $x = 1$

$$2^n = T_0 + T_1 + T_2 + T_3 + \dots + T_n$$

$$2^n - 1 = T_1 + T_2 + T_3 + \dots + T_n$$

Sol.57. (a)

$$(1-x^2)^{20} \left(2 - x^2 - \frac{1}{x^2} \right)^{-5}$$

$$(1-x^2)^{20} \left(- \left(x - \frac{1}{x} \right)^2 \right)^{-5}$$

$$= \frac{(1-x^2)^{20}}{\left(x - \frac{1}{x} \right)^{10}} = - \frac{(x^2-1)^{20}}{\left(\frac{x^2-1}{x} \right)^{10}}$$

$$= -(x^2-1)^{10} x^{10}$$

In expansion of $(x^2-1)^{10}$ constant term is last term that is 1

So coefficient of x^{10} is -1 .

Sol.58. (a)

$$\frac{2^{120}}{7} = \frac{8^{40}}{7} = \frac{(7+1)^{40}}{7}$$

Constant term is remainder that is 1.

Sol.59. (b)

$$\left(mx + \frac{1}{x} \right)^n$$

4th term

$${}^nC_3 (mx)^{n-3} \cdot \left(\frac{1}{x} \right)^3 = \frac{5}{2}$$

$${}^nC_3 (m)^{n-3} \cdot (x)^{n-6} = \frac{5}{2}$$

Given term is constant term so $n-6=0$

So $n=6$

$${}^6C_3 (m)^3 = \frac{5}{2}$$

$$(m)^3 = \frac{1}{8} \Rightarrow m = \frac{1}{2}$$

So $mn=3$

Sol.60. (a)

$$(x+y)^{2n+1} (x-y)^{2n+1}$$

Power is odd so there will be 2 middle terms
1st at $(n+1)^{th}$ position and next at $(n+2)^{th}$ position

Sum of middle terms is given zero

$${}^{2n+1}C_n (x^2)^{2n+1-n} (-y^2)^n + {}^{2n+1}C_{n+1} (x^2)^{2n+1-n-1} (-y^2)^{n+1} = 0$$

$$\Rightarrow {}^{2n+1}C_n (x^2)^{n+1} (-y^2)^n + {}^{2n+1}C_{n+1} (x^2)^n (-y^2)^{n+1} = 0$$

$$\Rightarrow (x^2)^{n+1} (-y^2)^n + (x^2)^n (-y^2)^{n+1} = 0$$

$$\Rightarrow (x^2)^{n+1} (y^2)^n (-1)^n = -(x^2)^n (y^2)^{n+1} (-1)^{n+1}$$

$$\Rightarrow (x^2)^n x^2 (y^2)^n = (x^2)^n (y^2)^n y^2$$

$$\Rightarrow x^2 = y^2$$

$$\Rightarrow \frac{x^2}{y^2} = 1$$

Sol.61. (d)

$$(8+3\sqrt{7})^{20} = U + V$$

$$(8-3\sqrt{7})^{20} = W$$

by adding both equations

$$U + V + W$$

$$= (8+3\sqrt{7})^{20} + (8-3\sqrt{7})^{20} = \text{Integer}$$

All irrational terms are cancelled out.

U is given in question so $V+W$ must be integer.

Now by rationalize W

$$W = \left(8-3\sqrt{7} \times \frac{8+3\sqrt{7}}{8+3\sqrt{7}} \right)^{20} = \left(\frac{1}{8+3\sqrt{7}} \right)^{20}$$

by above equation $0 < W < 1$

and given that $0 < V < 1$

by adding above two inequalities

$$0 < V + W < 2$$

$$\text{so } V + W = 1 \quad (\text{because } V + W = \text{Integer})$$

Sol.62. (b)

$$(8+3\sqrt{7})^{20} = U + V$$

$$(8-3\sqrt{7})^{20} = W$$

multiply both equations

$$(U+V)W = (8+3\sqrt{7})^{20}(8-3\sqrt{7})^{20} = 1$$

Sol.63. (b)

$$7^n - 6n$$

$$(6+1)^n - 6n$$

$$[{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \dots + {}^nC_{n-2} 6^2 + {}^nC_{n-1} 6^1 + {}^nC_n] - 6n$$

$$= [{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \dots + {}^nC_{n-2} 6^2 + 6n + 1] - 6n$$

$$= [{}^nC_0 6^n + {}^nC_1 6^{n-1} + {}^nC_2 6^{n-2} + \dots + {}^nC_{n-2} 6^2 + 1]$$

$$= 36[{}^nC_0 6^{n-2} + {}^nC_1 6^{n-3} + \dots + {}^nC_{n-2}] + 1$$

by dividing 36 there is 1 remainder.

Sol.64. (c)

$$(1+x)^p (1+x)^q = (1+x)^{p+q}$$

Let x^3 exists its $(r+1)^{th}$ term

that is General term,

$$T_{r+1} = {}^nC_r (a)^{n-r} (b)^r \text{ in } (a+b)^n$$

$$= {}^{p+q}C_r x^r$$

for coeff. of x^3 , $r=3$

so coeff. of x^3

$${}^{p+q}C_3 = 35$$

$$p+q=7$$

Sol.65. (b)

General term of $\left(\frac{1}{3^2} + 5^{\frac{1}{4}} \right)^{12}$ is

$${}^{12}C_r \left(\frac{1}{3^2} \right)^{12-r} \left(5^{\frac{1}{4}} \right)^r$$

By above expression, if r is multiple of 4, then there will be no rational term.

So $r=0, 4, 8, 12$ i.e. 1st, 5th, 9th, 13th terms are rational terms.

Sol.66. (c)

Sum of binomial coefficients $= 2^n = 256$

So $n=8$

Greatest term lies at middle position, and its

$$\text{middle term is } \left(\frac{8}{2} + 1 \right)^{th} = 5^{th} \text{ term}$$

Sol.67. (b)

$$k < (\sqrt{2}+1)^3 < k+2$$

$$k < 2\sqrt{2} + 6 + 3\sqrt{2} + 1 < k+2$$

$$k < 5\sqrt{2} + 7 < k+2$$

$$k < 14.07 < k+2$$

$$k < 14.07 \text{ and } k > 12.07 \text{ so } k=13$$

Sol.68. (c)

$$5^{99} = 5 \times 5^{98} = 5 \times 25^{49} = 5 \times (26-1)^{49}$$

Each term of expansion of $(26-1)^{49}$ contains 26, except only last term ${}^{49}C_{49}(-1)^{49}$,

So last term will give remainder -1 , and 5 is given in multiply so remainder will be -5 .

But remainder cannot be negative, when remainder comes negative than we add divisor in remainder so correct remainder is $-5 + 13 = 8$

Sol.69. (d)

Greatest coefficient exists at middle term, and

middle term of expansion of $(a+b)^{100}$ is 51th

term and this term is

$$T_{51} = {}^{100}C_{50} (x)^{50} \text{ for } (1+x)^{100}$$

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5 Youtube video:

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